RESULTS

For the majority of us, *Agaricus bisporus* is the type of industrial fungus most commonly found on pizza and available in grocery shops. It comes in a variety of shapes, including button, brown, and big Porto variations, all of which belong to the same species. Its popularity is more a result of its adaptability for industrial use than of the flavor as a whole. However, all of the *Agaricus* mushrooms sold in stores are cultivated on a controlled media and in a controlled environment, sometimes in caves or underground constructions, not so much for the darkness but because of the constant temperatures and high humidities. *Agaricus* does grow wild, typically in fields or lawns.

A thorough analysis outlining the nutritional, pharmacological, and cosmetic benefits of *A. bisporus* as well as identifying the research gaps is offered to emphasis various elements of the plant. By doing this, it would be feasible to increase the number and quality of bioactive chemicals in *A. bisporus*, which might eventually aid in the identification of novel therapeutics and the underlying mechanisms. We highlight the most recent discoveries about the dietary, medicinal, and aesthetic benefits of *A. bisporus* in this article. Research gaps and potential future directions are also highlighted.

In terms of understanding the gastronomic and therapeutic benefits of mushrooms, *A.bisporus* has broadened the world's perspective. Tyrosinase and ergothioneine have been isolated from this fungus in recent years, making it more valuable to consider for dietary and therapeutic applications. Agaricus and almost all other fungi that produce stalked structures with a cap, or that would be classified as mushrooms, are club fungi, also known as basidiomycete fungi (Phylum Basidiomycota). The majority of mushrooms have 'gills' where spores are formed on the underside of the cap, and Agaricus exhibits this characteristic.

Like most creatures, *Agaricus* depends heavily on its interactions with other organisms and its physical surroundings to survive. The websites below that discuss how mushrooms are raised commercially reflect this. *Agaricus*, as previously mentioned, is referred to as a "secondary decomposer" since it consumes material that has already been consumed by "primary decomposers." This is comparable to how cows and the microbes in their stomachs interact. Even while the order Agaricales, which bears *Agaricus'* name, still exists, it does not include all gilled mushrooms and does include a few fungi that do not. A growing number of byproducts from their industrial production are a direct result of *Agaricus bisporus*, which constitutes a significant food sector in Europe.

Preparation of compost

First of all we have to prepare the compost. Compost is basically prepared with the help of animal manure and wheat straw. They are mixed well with the 1.1 ratio and the after few times later the spawn is inserted in it. This preparation takes few weeks in which proper water, air and care is important for proper and well compost. It starts with wheat straw collection. It is taken out from the collection in a separate plate and placed in a vessel having boiling water. This is done so because to sterilize wheat straw.

Fig. 1. **A** is showing the sample of wheat straw that is taken from the collection This will be boiled for the sterilization purpose in a separate vessel

B is showing the total collection of wheat straw present in a carton box. According to our needs, we take the sample out of the collection

Sterilization process

First of all water is taken in a vessel and boiled at high flame. The purpose is to get the boiling point of water. When it is done, Wheat straw is taken in a separate vessel and added to that boiled water and mixed continuously with the help of a stirrer so that there should be proper mixing and sterilization process. This is clearly shown in the diagram below.

Fig. 2. C is showing that the sample of wheat straw is taken and is added to the boiling water. The purpose of all this is to sterilize it.

D is showing that loading of wheat straw is done in the vessel

Fig. 3. A is showing that the wheat straw is added and well mixing is being done with the help of stirrer so that there should be proper mixing up. The flame is high which means the constant heating and mixing is necessary for proper results.

B is showing that mixing and stirring is done. It is time to take these wheat straw out of that boiling water with the help of strainer and it should be done with proper care.

Fig. 4. A is showing that the sample of wheat straw is taken out of boiling water and the place where it has to be placed is cleaned properly and is taken on the newspaper to avoid from germs.

B is showing the manure sample is taken in a separate vessel. This manure is also vital in the preparation of compost.

Well mixing of wheat straw and manure

Fig. 5. A is showing that compost is being prepared with the help of wheat straw. Equal amount of manure and wheat straw is taken in a separate vessel with ratio of 1:1

B is showing that after the equal samples are taken there is well mixing of the compost with the help of protected and safe hands

Vessels to keep that compost

There are different kinds of vessel that are used to keep that compost having spawn and manure. Here we are going to take a large sized bucket having holes so that there should be proper aeration and trays of different types are taken. Different cleaned and sterilized newspapers are also taken with can help to cover that compost to avoid any kind of dirt or germs etc.

Fig. 6. A is showing a vessel commonly a bucket having the different kinds of holes on the front side for the purpose of proper aeration

B is showing the tray where we have to insert the compost and the spawn for proper growth. This should be neat and clean.

Organized Compost

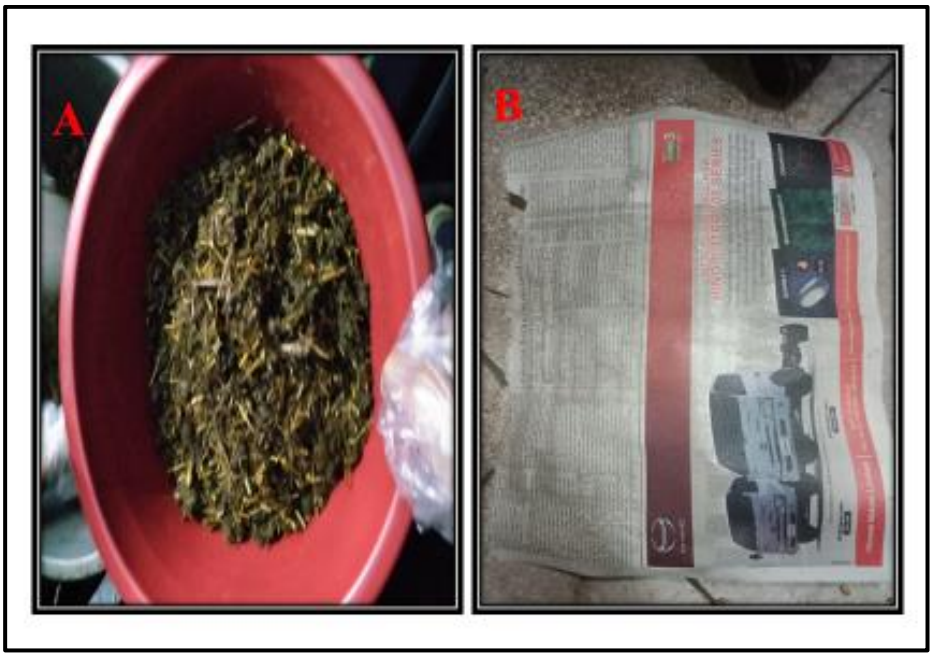

Fig. 7. A is showing that the compost is ready and taken in a bucket having holes for proper stay and we have to keep that until the growth of spawn starts

B is showing that the compost is taken in a tray that is covered the newspaper

Growth of mushroom

Fig. 8. A is showing that the growth of the mushroom has been started and s small part of this can be shown in the diagram

B is showing that the white coloured portions have spawn of mushroom that are ready to grow

White button mushroom

White button mushroom (*Agaricus bisporus)* is grown. *Agaricus* does grow wild, typically in fields or lawns but all of the *Agaricus* mushrooms sold in stores are grown on a controlled medium and in a controlled environment, often in caves or underground structures, not so much for the darkness but because of constant temperatures and high humidity. *Agaricus*, and nearly all of the fungi that would be described as mushrooms, i.e. that produce stalked structures with a cap, are club fungi = basidiomycete fungi (Phylum Basidiomycota).

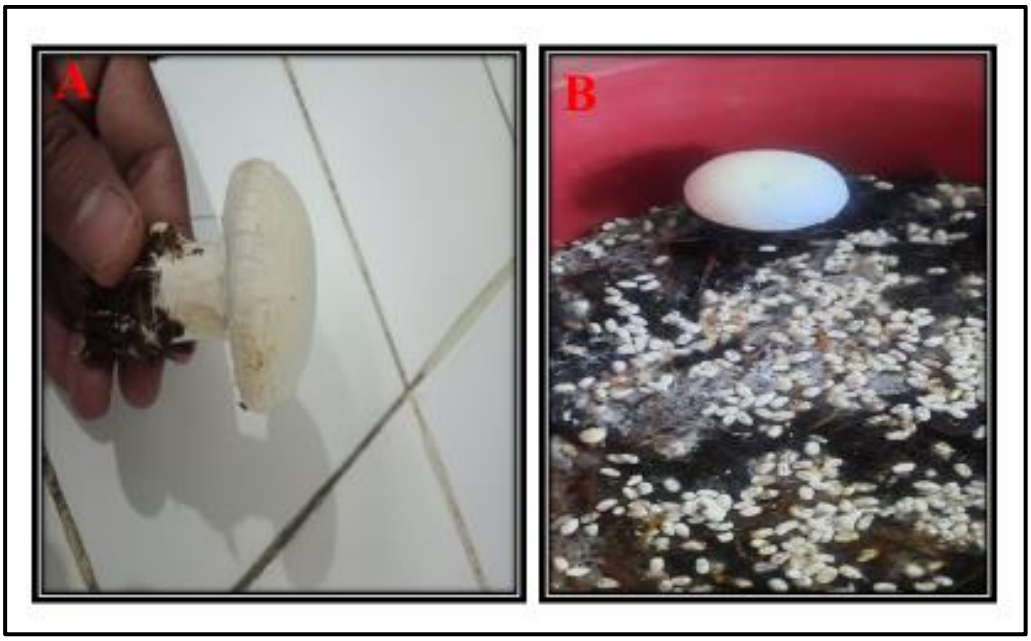

Fig. 9. A is showing that finally the white button mushroom (*Agaricus bisporus)* is grown. The complete shape is clear having cap, stalk and roots etc.

B is showing that an obvious button mushroom (*Agaricus bisporus)* is present and well grown in a vessel along with other spawn samples.

Gel electrophoresis

The process of gel electrophoresis is done for the purpose of getting the bands. This process was done again and again and finally a band of 50KDa was obtained

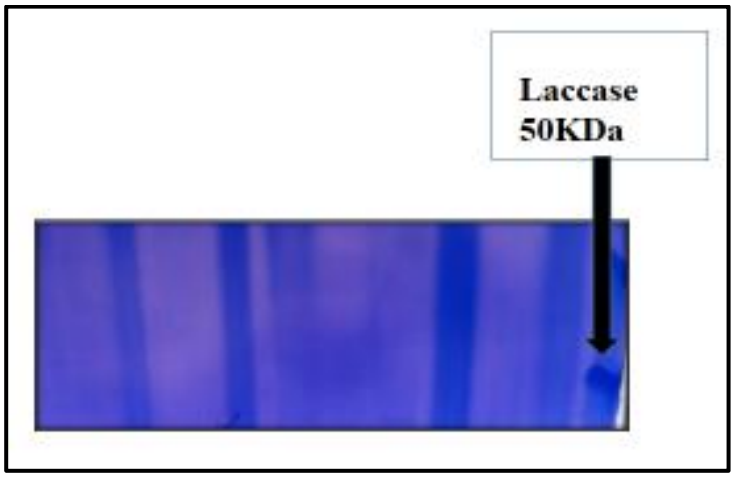

Fig. 10. The process of gel electrophoresis is done for the purpose of getting the bands. This process was done again and again and finally a band of 50KDa was obtained on 05-03-2023 that can be seen with the help of diagram shown above.

Agaricus bisporus secretes abundant laccase activity into the medium during mycelial growth. SDS-PAGE analysis of extracellular laccase protein, purified from compost extract, showed a predominant band of 50 KDa molecular mass, together with lesser amounts of smaller polypeptides. The main polypeptide was purified electrophoretically. Amino acid sequence analysis of the N-terminal region of the main polypeptide was used to specify the sequence of a 15-residue chemically synthesized peptide (N-terminal peptide). Rabbit antibodies were raised against pure laccase, electrophoretically purified main polypeptide and the synthetic N-terminal peptide. Electrophoretically purified main polypeptide antibody was further purified by affinity chromatography on laccase-CNBr-Sepharose. Western blot analysis showed that the antigenic behaviour of laccase in compost extract, culture filtrate from malt-extract culture, and the purified enzyme from both sources, differed.

FASTA Sequences of AbL

FASTA sequence of *Agaricus bisporus* laccase was retrieved from NCBI database (Accession No ACC18877).

CTACGTCTTGACAGTATGTCTCTGAAGCACATTGTTAATCATGAGCGAACCTAAG
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AGTATTAAATCACAAACTGAACCACAGTACAACGCCTGCAAGACACGCGCGTGA
TGTTCAATACAAACGTTGAACATCTATGCGCCTGTAACGGGGAGAGAAACCTCG
TGAAACTGCCCTGGATGATAAATTTCATCAGCTTCGAATGTCAGTGACATTGAAT
GTTATAAATACCCAGCAGAGAGTGGCTCGTCCTCAATGTGAGCGCGTAAACTTTA
GAAAATTTCCGTGGATTGTGGTCTCAGGCATATCGCTGGTGGTTTCTCCTTGTTGA
TAAAAATATGTATCTTCAGGCTATAAAGACGGGGGCAGAGGACCGTGGATTGAC
TCTCCAGCGGTAGGATCAACAATGAGGCTGTCCAACGCTTTAGTATTGGTCGCCG
CATGCATTTCGAGTGTAGTTGCGAAAACCAGAACCTTCGACTTCGACCTAGTCAA
TACCAGGCTCGCGCCTGATGGGTTTGAAAGGGAGTGAGTATATTTCTATACAGAT
TCATCGTTAGAGTTTACTGACCTTGATTTAAACTAGCACCGTGGTCATCAATGGA
GAATTCCCTGGAACACTGATCCAAGTCAACAAGGGCGACAGTGTCCGCATCCCT
CTCCATAACAAACTGACTAGCCCGACCATGAGGCGAAGTGTATCCATTGTGAGTT
ACTTTTTGAGCCCAGATTGACTTGACTATATAACACAAATTAGCATTGGCATGGA
TTTTTCCAAGCGAGAACTTCTGGTCAGGACGGGCCTTCCTTCGTGAATCAGTGCC
CTCAGCCACCCAACACAACCTTCACCTATGAATTCAGCGTGGCGGAGCAGTCTGG
AACGTTCGTCCTGATCCCTTAAATACATCTTCAGCGTAAATTGACATTTCTTCAAG
TTATTGGTACCATTCACACTTGTCCACCCAATACTGTGATGGTCTTCGCGGTGCAT
TCATCGTTTACGGTAATTCCTTCCTCTTGCATGCGAAGATATTTTTGCGACTTTCG
TTTAGACCCACGTGATCCACTCAGACATCTCTATGATGTTGATGACAAGTACCGT
CATCACTCTAGCTGAATGGTATCACATCCTCGCACCCGATGCCACTACGAGTTTT
TCAGCTCTGGCATTATACCGTAAGCTTCTTAAAGGAATATTTATTCATTTTCATAA
CCCAATTCCTGCAGCGTGCAAGATTCTGGCTTGATCAACGGTAAGGTAGATTCAA
TGGAGGACCCCTCAGTAAGTAGATCAAAACAGAAAAGGTGATGATGAAAATTTG
ACTTTACTTGCTTAGCACCATTCGCCGTCGTCAATGTTCGCCGAGGAAACGTTAC
AGGCTCCGTGTCATAGCTATTTCTTGCAGGCCGTTCTTCACCTTCTCCGTCGATAA
CCATAGCTTGGTCTTCATGGAAGCCGATGGTGTTGAACATGACCCTGTCGAAGTC
CAAAACGTCGACATCTACGCGGCCCAGCGTGTCTCGGTTATTCTGCATGCCAATC
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GGAACCCCAATCGTAAGCGTCATTTTGCCACCTGCTAGCAGCTGTAATGAAACGT
GTTTCCAAAGTAAACATCAGCCTCACTCTCGCCATCCTGCGCTACCATGGTGCAC
GTCACGTCGAACCGACTACGGTCAACGTTCCCGGACACAAACTCCTTGATCAAG
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ACCAATCTAGCCGATCCGCCAGGAAGGACCAGGCAAAGTGTGTATATATTTACA
TGAACATAATCATCAACTCAACGTTCTCTAGCTCGGTGACGGTCCCCCCGATAAG
CACATCACCCTCAACATTGCTCAAGTAAGCGTCTCATTATTCAAATCAAACTCGA
CTTCTGACATGAGCACAGCCCAACGCTCCCTTCTTCGATATCAACGGAATCTCTT
ACATCTCCCCGACTGTCCCTGTTCTTCTCCAAATCCTCAGCGGAGCCAAGAGGCC
CGAAGATGTTCTTCCATCCGAACAAATCTTCTTTGTCCCCAAGAATTCATTGATTG
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TTCTCCTCAATTCCTCTACTTACGTCCCTGTTTCAGTGCACGGTCATAACTTTGAT
GTCGTTCTGGCTTCTAACGACGACACTTTCAACTTCAAGAAGTGAGTTAATTGAA
AAAGGAAGTTACCTGCTGAACACTAATCTCAAATGAAAGCCCGCCTCGTCGTGA
CGTATACCCCATCAACGGTGGCAACACCACTTTCCGCTTCTGTAAGATTCCACGT
TTCTTTCTGAAGAATAACTCTTTCTGAACGACCCCTCAGTCACCGATAACCCAGG
CGCATGGTTCTTGCATTGGTGAGGAACATTTCTATCCATTTCTCTCGAGGCGAAT
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TACAAGTAGGAATTTTAAACAATGTTTACCCTAAAACAAAATGATCGTAAGTATT
AAGTGGCGGCATTAACGAGGCAGGTAGTAATGATAACAAACCCAGGCACTTGAT
GCGCTTTGAATTGAAGTCTTAAGGGCCCATGTGTAACAAGAATGCATTCACTGAC
ACGACCATGAGACTCCGAACGGTCTATATGGCACTGTCACAAGGTGATGAGTGA
AGATAGAACAAGTTTACCCAGATCTGCTGAAGTTGCTGATCCTGGCAGACTGTCG
ATTCTCACGGAGCCTTAAATTCTTCCCATGCTCATTGGCATCAGTTCCCACTGTCA
CTCGCACGATTCACACATGCGAGATGACATTGTTGAGCGTATGAAAGTACTTCCA
GGAAGTAATATAAATCATATTAGCGAGATAATATTGACAATGTTTGGGAAATTTT
CTCATAGTGAGTTTCACATCCGGCTCTAACTGAAGATAGGCAGGTACTCCAATGA
GATGCCTCCAAATTACCTTCAATCATCCAGTCCAAATTGCACTGCAAAGTATTTT
GCGCTTCCACGGCACGTCCCTCAACAATATACTAAATAAATGATTCCTTAATGAT
TCGGCGGTATGGCCGCCTTATTGAAGGTCGGCGGCGCATTGGAATCTTCAATTTT
GGCAATCTATATGGTCCACGCCTCTCCGTGCTTACTGCTTAGTGAAGCGGCGAAG

The FASTA sequence of AbL enzyme was retrieved from the translated protein sequence from that gene (NCBI Accession No. ACC 18877). This sequence can also be retrieved from UniProt ID: Q12541 in the UniProtKB database. Both sequences were the same with Taxonomic identifier in NCBI.

MRFSNAFVLVAACISSVLADTKTFNFDLVNTRLAPDGFERDTVVINGEFPGTLVQVN
KGDSVRIPVNNKLTSSTMRRSVSIHWHGFFQARTSGQDGPAFVNQCPQPPNTTFTYE
FSVADESGTFWYHSHLSTQYCDGLRGAFVVYDPEDPLGHLYDVDDETTVITLAEWY
HVLAPDINNEFFSSGIIPVQDSGLINGKGRFNGGPETPFAVVNVEQGKRYRFRVIAISC
RPFFTFSVDNHNLTFMEADSVEHDPVEIQNVDIYAAQRVSVILNANQPVDNYWMRA
PMTGGNPDRNPNLNISLTLAILRYKGAPEVEPTTVNVPGHKLLDQEMHPIAQEGPGK
LGDGPPDKHITLNIAQPNAPFFDINGISYISPTVPVLLQILSGAKRPEDVLPSEQIFFVPK
NSLIEVNIPGEGAHPFHLHGHNFDVVLASNDDTFNFVNPPRRDVYPINGGNTTFRFFT
DNPGAWFLHCHIDWHLEAGLAIVFAEAPEDNVSGPQSQITPQDWLDLCPEYNAIEPE

Homology Table

Four possible templates PDB ID: 3FPX, PDB ID: 3PXL, PDB ID: 5MHU and PDB ID: 5A7E) were selected by homology sequence search. All these chosen templates had query coverage greater than 98% and percentage identity greater than 98%.

Sequence homology percentage identity matrix was observed, and a homology table (Table 1) was built (https://www.ebi.ac.uk/Tools/services/rest/clustalo/result/clustalo-I20220816-142744-0204-91031055-p1m/pim) after the multiple sequence alignment in Clustal Omega server.

Table 1. Sequence homology table between the AbL enzyme with possible templates shows that the enzyme is much similar to template.

Sr. No.	Enzymes	Homologies (%)
1	ACC1	100.00
2	5MHU	49.28
3	5A7E	48.66
4	3PXL	48.87
5	3FPX	48.87

Multiple Sequence Alignment

For the proper understanding of the comparative relationship among all these structures (mentioned in Table 1), Multiple Sequence Alignment (MSA) was done on the Clustal Omega online server, and then from the ESPript 3 server, the similarity MSA (Fig. 1) was obtained. This is shown in the below-given diagram.

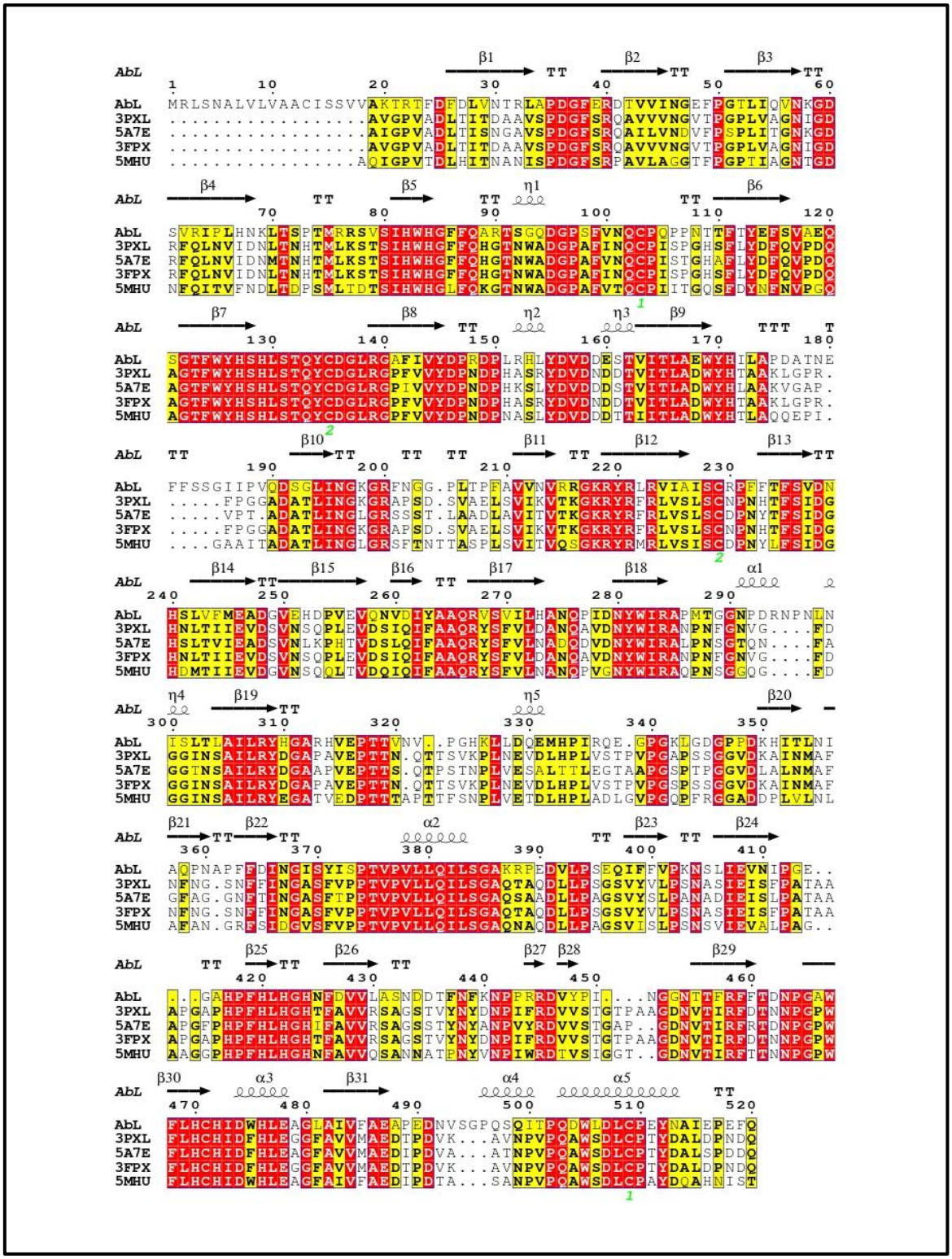


Fig. 1. MSA of AbL and templates was drawn by ESPript 3 server. Identical and similar residues are shown in red and yellow boxes, respectively. Secondary structures of AbL are displayed at the top of aligned sequences. Helices are represented in squiggles, β- strands in arrows, and turns are indicated in TT letters.

Phylogenetic Tree

A phylogenetic tree was constructed (Fig. 2) to see the evolutionary taxa of these AbL and its possible templates and to select the most suitable template. The tree showed that 3PXL is the nearest and best template for structure prediction among them. Although data are becoming more numerous and strong analytic tools are accessible, there are several barriers to trustworthy tree construction.

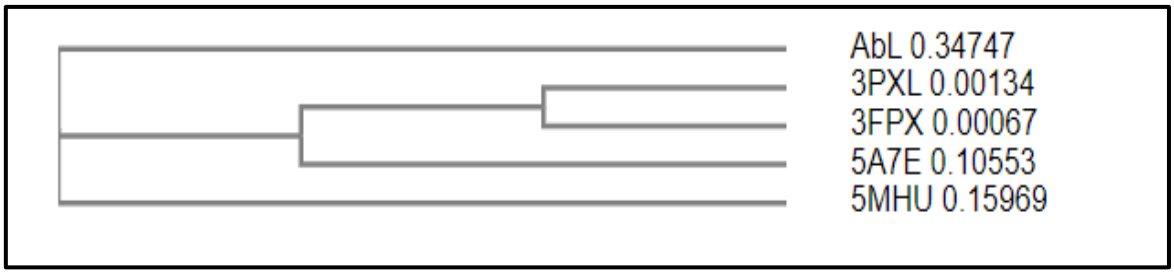
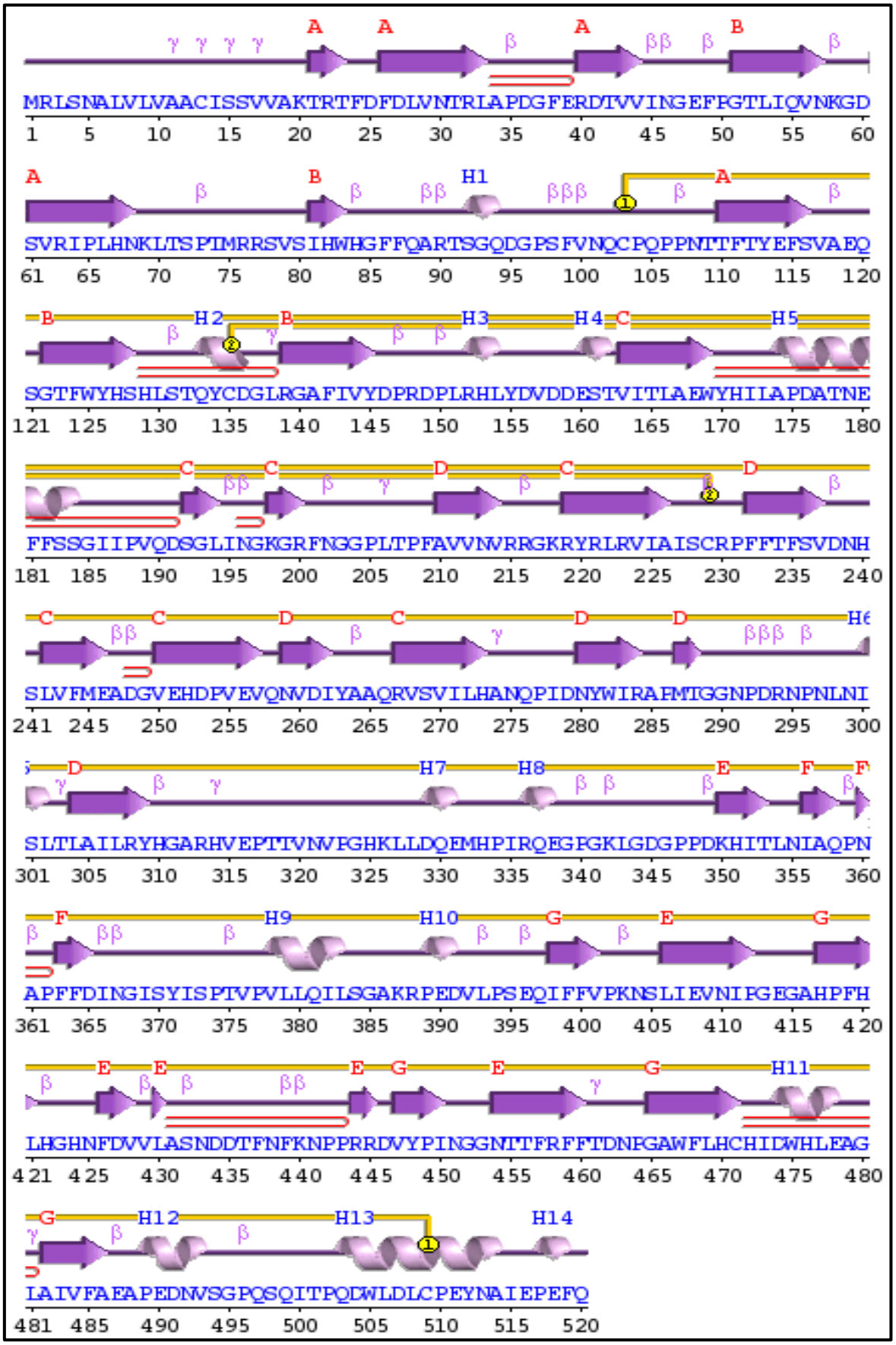


Fig. 2. Phylogenetic tree of AbL and templates represents that PDB ID: 3PXL is the closest relative of AbL as compared to other templates. It also indicates that it can be the best template for 3D structure prediction of AbL

Secondary Structure of AbL

AbL has a molecular weight of 50.064 KDa, and it contains a total of 520 amino acids. Secondary structural conformations of the predicted model of AbL were generated.


MRLSNALVLVAACISSVVAKTRTFDFDLVNTRLAPDGFERDTVVINGEFPGTLIQVNKGD
1 5 10 15 20 25 30 35 40 45 50 55 60
SVRIPLHNKLTSPTMRRSVSIHWHGFFQARTSGQDGPSFVNQCPQPPNTTFTYEFSVAEQ
61 65 70 75 80 85 90 95 100 105 110 115 120
SGTFWYHSHLSTQYCDGLRGAFIVYDPRDPLRHLYDVDDESTVITLAEWYHILAPDATNE
121 125 130 135 140 145 150 155 160 165 170 175 180
FFSSGIIPVQDSGLINGKGRFNGGPLTPFAVVNVRRGKRYRLRVIAISCRPFFTFSVDNH
181 185 190 195 200 205 210 215 220 225 230 235 240
SLVFMEADGVEHDPVEVQNVDIYAAQRVSVILHANQPIDNYWIRAFMTGGNPDRNPNLNI
241 245 250 255 260 265 270 275 280 285 290 295 300
SLTLAILRYHGARHVEPTTVNVFGHKLLDQFMHPIRQEGFGKLGDGPPDKHITLNIAQPN
301 305 310 315 320 325 330 335 340 345 350 355 360
APFFDINGISYISPTVPVLLQILSGAKRPEDVLPSEQIFFVPKNSLIEVNIFGEGAHPFH
361 365 370 375 380 385 390 395 400 405 410 415 420
LHGHNFDVVLASNDDTFNFKNPPRRDVYPINGGNTTFRFFTDNFGAWFLHCHIDWHLEAG
421 425 430 435 440 445 450 455 460 465 470 475 480
LAIVFAEAPEDNVSGPQSQITPQDWLDLCPEYNAIEPEFQ
481 485 490 495 500 505 510 515 520

Sec. struc: Helices labelled H1, H2, ... and strands by their sheets A, B, ...
Helix Strand

Motifs: β beta turn γ gamma turn beta hairpin

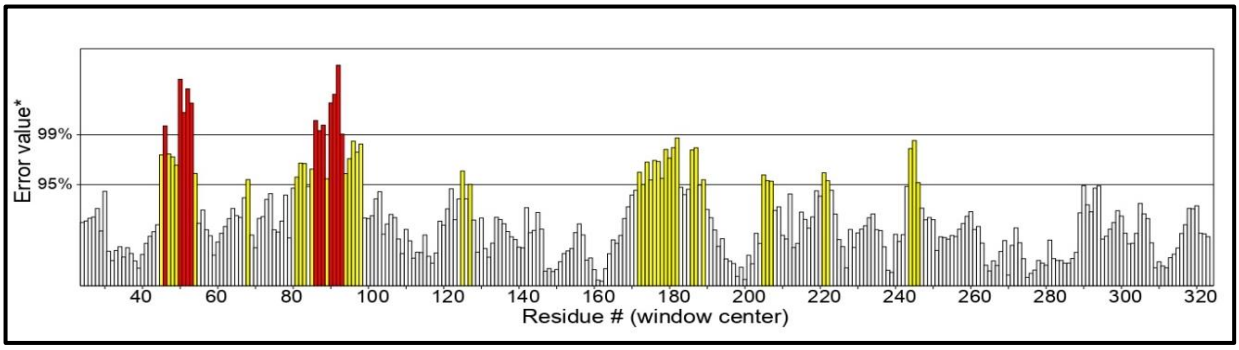
Disulphides: disulphide bond

Residue contacts: to metal

Fig. 4. Secondary structure of AbL represents the helices in purple, which are H1-H18; beta sheets are shown in yellow arrows, and loops are indicated in brown lines. Domains of AbL are shown in rod-shaped lines above the primary sequences, C-terminal in domain 1 (residues 1 to 18 and 232 to 418) in blue color and N-terminal in domain 2 (residues 22 to 228). Residues in the pink background show the double-stranded linker peptides

ERRAT for AbL

The ERRAT quality factor was found to be 86.558%. (Fig. 5). ERRAT shows the proportion of interactions of non-bonded atoms in the structure. The quality score >90% shows the most reliable structures and is produced mainly by better quality structures. About >90% quality factor (produced by lower resolutions 2.5-3.0 Å) is acceptable to consider the structure



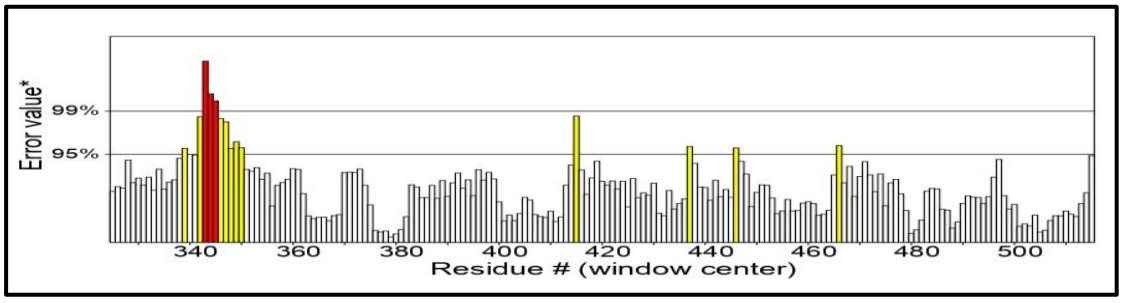


Fig5. ERRAT shows an overall quality factor of 86.5%. On the error value axis, two lines (yellow and red) are drawn, indicating the residues disturbing the structure's overall quality. These show the confidence with which it is possible to reject regions that will exceed that error value. The more the quality factor of the structure, the more reliable the structure is.

Ramachandran Plot for AbL

Ramachandran plot was obtained from the PROCHECK to evaluate backbone conformations and stereochemical quality of the modeled structure by Φ-ψ plot. In a Ramachandran plot, the white regions represent conformations where atoms in the polypeptide approach are closer than the total of their van der Waals radius. These areas are sterically forbidden for all amino acids except glycine, which is unusual because it lacks a side chains The red sections correspond to conformations where there are no steric conflicts, i.e., these are the authorized regions, notably the alpha-helical and beta-sheet conformations. The yellow areas illustrate the permissible regions if slightly shorter van der Waals radii are employed in the computation, i.e., the atoms are allowed to move a bit closer together. This brings out an extra area that corresponds to the left-handed alpha-helix.

Ramachandran plot for AbL (Fig. 6) revealed that 85.9% with 359 residues are present in the most favored regions, 12.4% with 52 residues in the additional allowed regions, and 1.2% residues are present in the generously allowed and the disallowed areas. Total residues are 503, no. of non-glycine and non-proline residues are 418, no. of end residues (excl. Glycine and Proline) are 2, no. of glycine residues are 36 and no. of proline residues are 44. A good quality structure has more than 90% residues in the most favored regions.

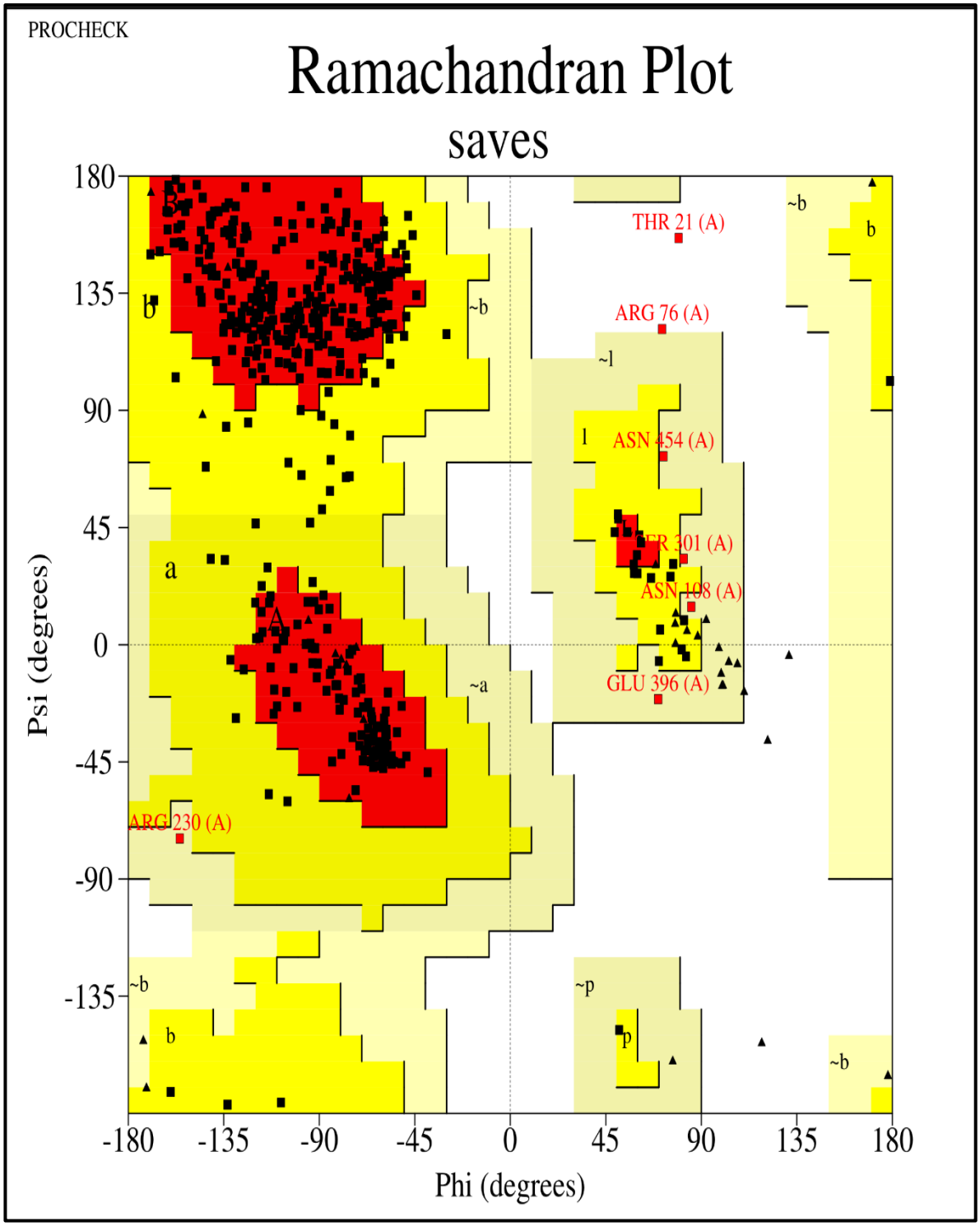


Fig. 6. Ramachandran plot for AbL. Regions [A, B, L] show the most favored regions, [a, b, l, p] indicate the additional allowed regions, [~a, ~b, ~l, ~p] show the generously allowed regions, while the remaining white region shows the disallowed regions.

VERIFY 3D

VERIFY 3D results showed that 83.20% of the structure residues have averaged 3D-1D score >= 0.1. At least 80% of the amino acids have scored >=0.1 in the 3D/1D profile.

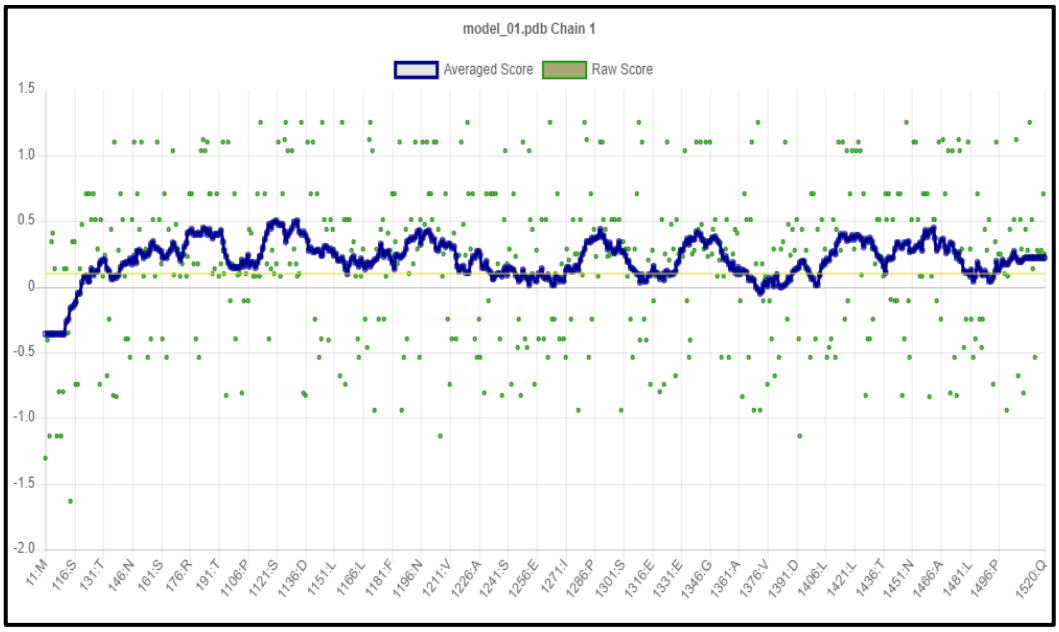


Fig. 7. VERIFY 3D for AbL indicated that 83.20% of all residues had averaged 3D-1D scores greater than 0.1. It represents that the structure is highly reliable

Structure Prediction

Predicted model of AbL

From the SWISS-MODEL templates generated for AbL enzyme, from *Agaricus bisporus* (PDB ID: 3PXL) was selected as a template because of its high coverage, high sequence identity (46.63%), high resolution (X-ray, 1.2Å) from PDB, and high Global Model Quality Estimate (GMQE) = 0.81 and Quaternary Structure Quality Estimate (QSQE) = 0.54 values as compared to other templates and 3D model of AbL was generated.

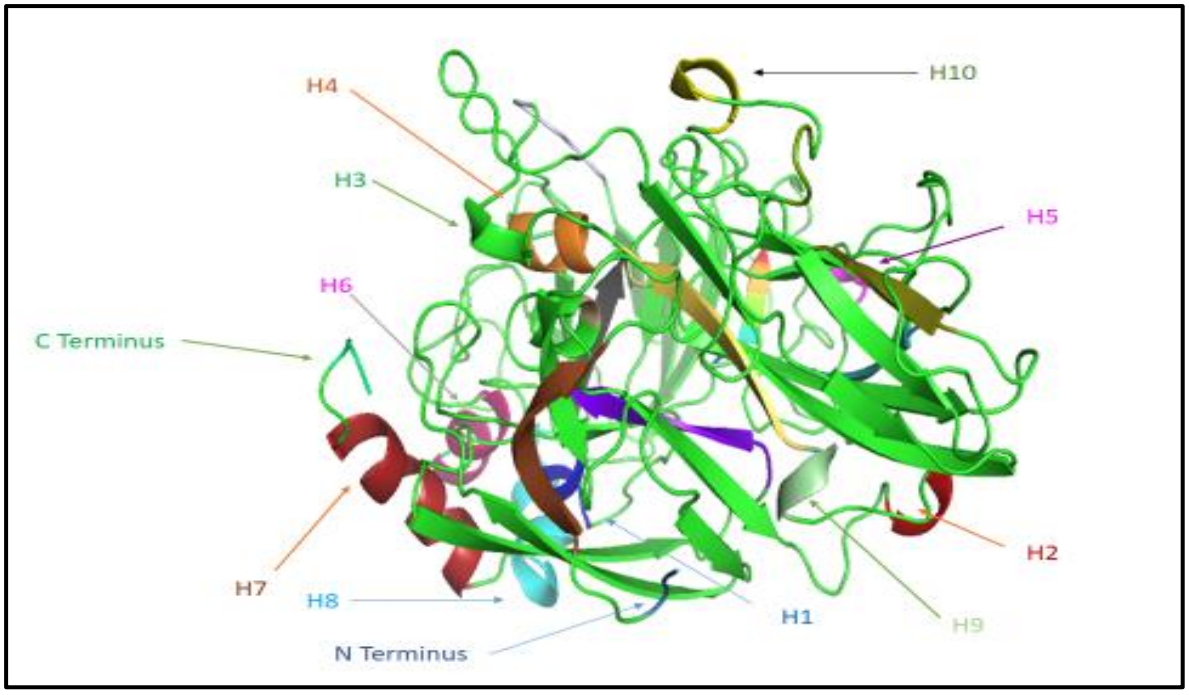


Fig. 3. 3D predicted model of AbL shows that it is composed of 2 domains shown in different colors. The model is represented in a cartoon model with domain 1 in blue color and domain 2 in green color. A double-stranded linker peptide connects the two domains and is illustrated in pink, while the PEP binding region is indicated in orange.

Structural Alignments of AbL

Structural alignment of predicted AbL (Fig. 8) with its template (PDB ID:3PXL) was done on the PyMOL Visualization Software version 2.5.2, which showed that alpha/beta arrangement, three-dimensional structure, functional domain, and conserved residues are similar by the superimposition of AbL and RMSD value of 1.38 Å was obtained.

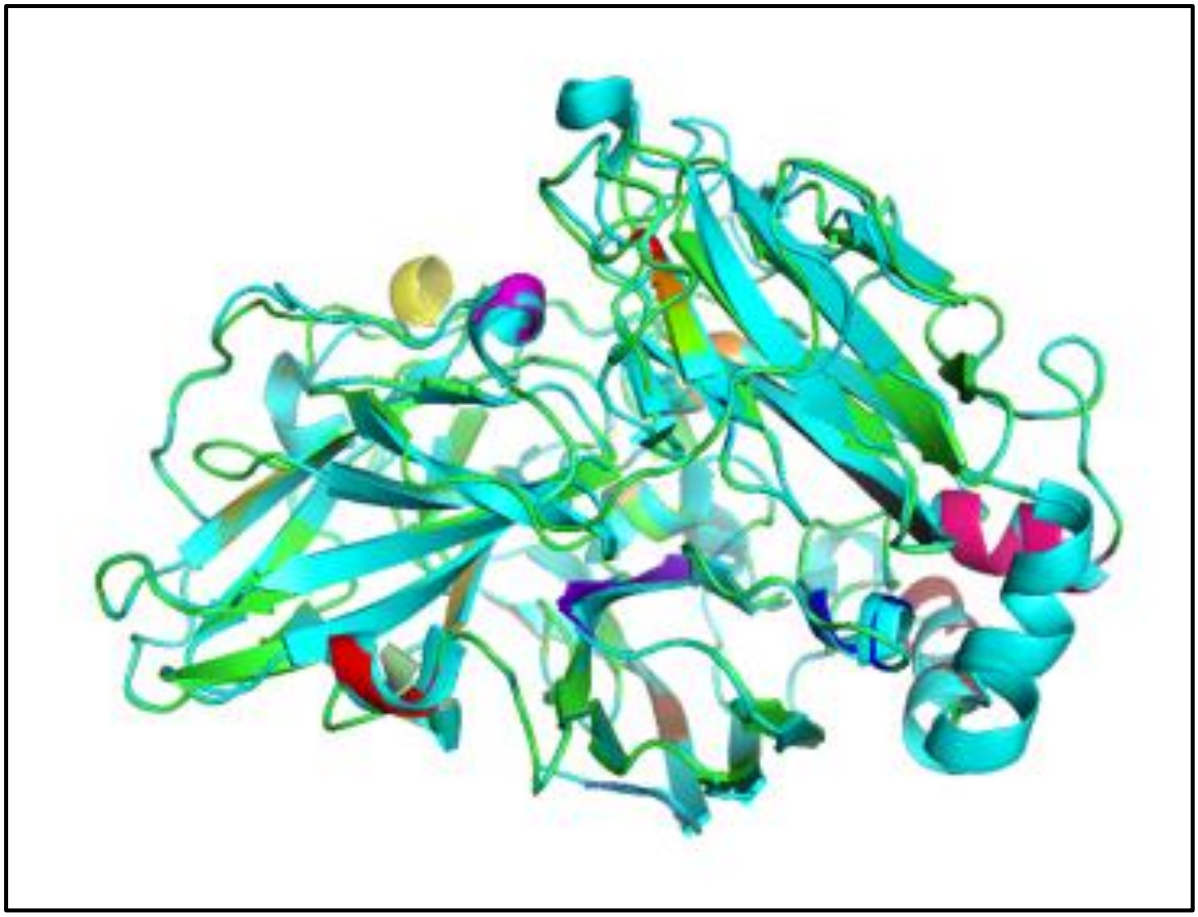


Fig. 8. Structural alignment of AbL with its template model showed the RMSD value of 1.38 Å. The template model is shown in blue color and AbL in green color. These validation results demonstrated that modeled 3D structure is exceptional and exhibits excellent structural integrity and stability. This evaluation and analysis conclude that the *Agaricus bisporus* AbL model generated in the current work is trustworthy for defining future protein-substrate, protein-ligand interactions and analyzing the link between the structure and function.

Molecular Docking of AbL and Glycerol

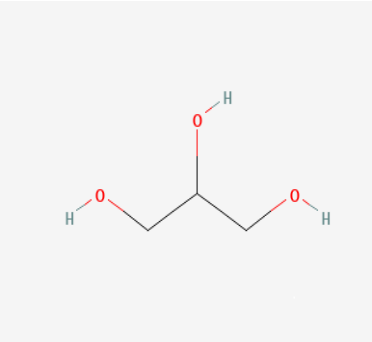
Obtain the Agaricus bisporus laccase 3D structure from the Protein Data Bank (PDB) or, in the absence of one, use homology modelling to predict the structure.Remove any ions, water molecules, or other tiny molecules that aren't a part of the enzyme to prepare the protein structure. Assign atom kinds and include any extra atoms required for a full structure. Build the 3D structure of glycerol using molecular modelling tools or obtain it from chemical databases. Glycerol should be converted to the proper 3D structure format, such as PDB, SDF,

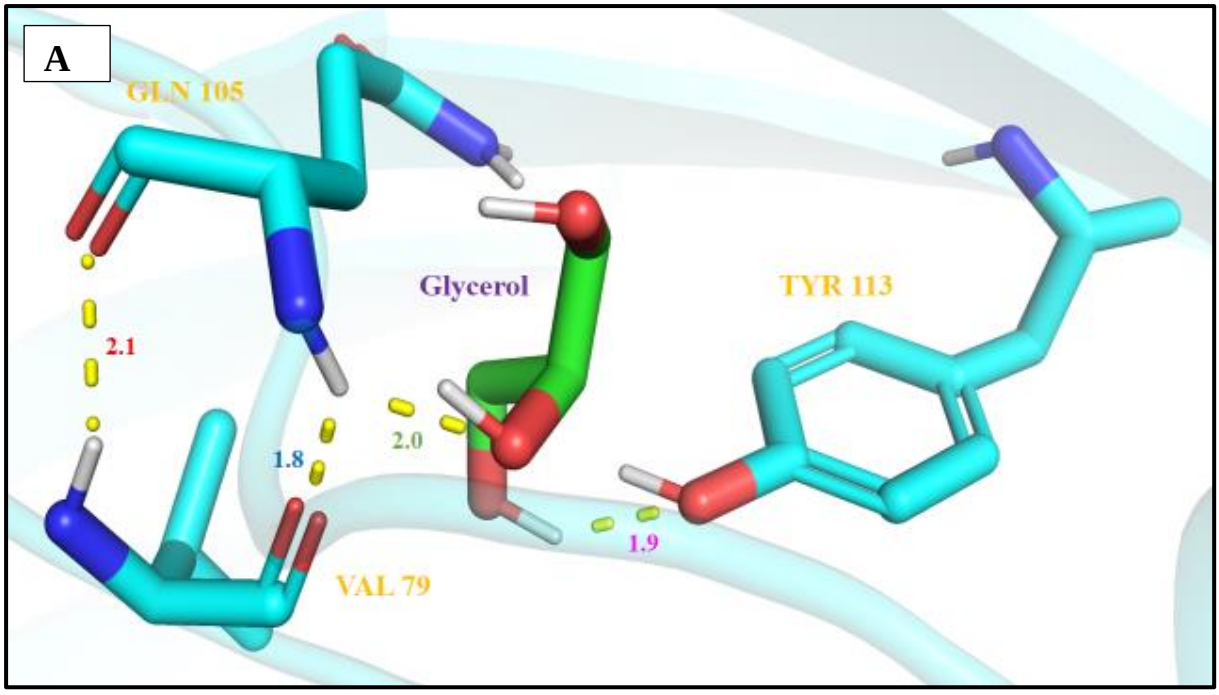
Molecular docking between AbL and Glycerol can help predict how Glycerol interacts with the enzyme's active siteand estimate the strength of their binding affinity

Molecular docking of AbL with the Glycerol was performed by using the binding pocket of the template (PDB ID:2QT6). It showed some residues of the AbL which interacted by forming H-bonds with ligand atoms (Fig. 9). These interacting residues were taken as the active binding site residues.

Fig. 9 represents that the GLU formed H-bonds with Val 79, Gln 105, and Tyr 113 with specific binding free energy (-6.7 kcal/mol) and RMSD value of 1.38 Å (Table 2). Molecular interactions between protein and GLU were analyzed via PyMOL-generated ball and stick protein model and LigPlot was generated with the help of PDBsum generate and downloaded

Table 2. Docking result analysis of natural substrate Glycerol and AbL indicating docking score, RMSD value, interacting residues with bond lengths, and 2D structure

PubChem ID	**Chemical 2D structure of Ligand**	**Docking Score (kcal/mol)**	**RMSD Å**	**Residues forming H-bonds**	**Distance Å**
GLU 753		-6.7	1.38	VAL79 GLN105 TYR113	2.51 2.83 2.73



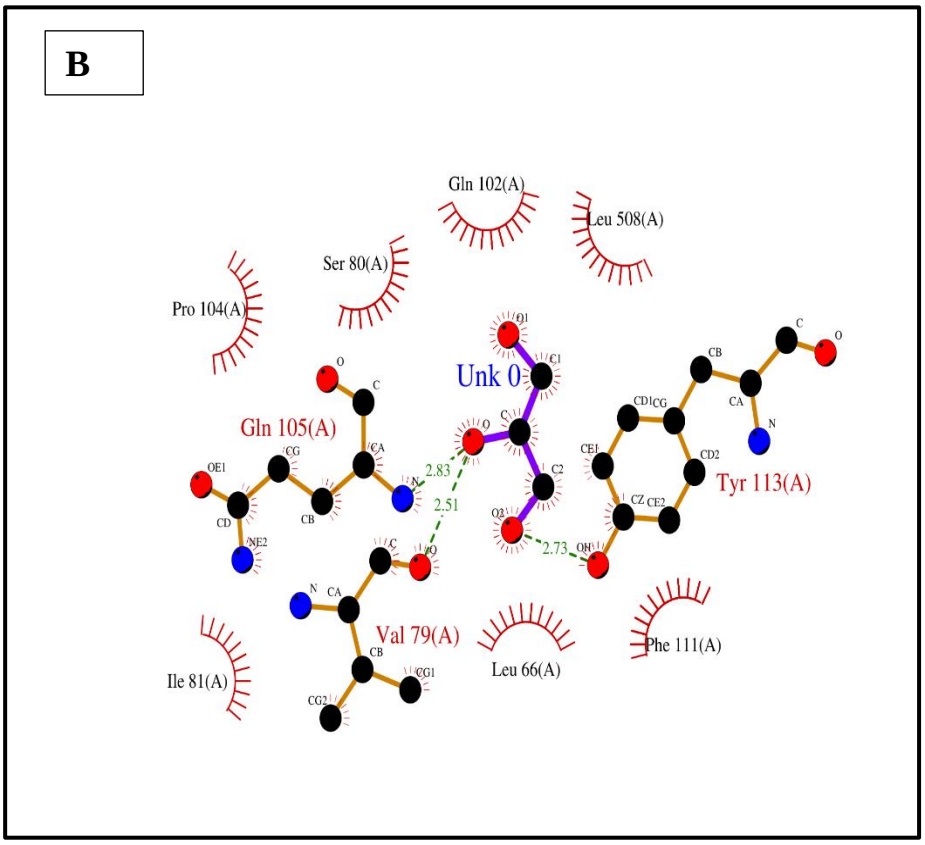


Fig. 9. Docking analysis of AbL GLU complex. (A) is showing the interactions in PyMOL, and (B) indicates the LigPlot analysis. It is evident that the ligand molecule makes 3 H-bonds (indicated by dashed lines) with the AbL active site residues (Val79, Gln105, and Tyr113).